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Aligning AWS native monitoring tools with Secure by Design

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Background

Monitoring is an essential capability for government organisations which use systems in the cloud. As applications become increasingly distributed and dynamic, visibility is fundamental to reliability, performance and security.

AWS provides a collection of native monitoring tools that allow organisations to:

- observe system behaviour
- detect deviations
- respond swiftly to emerging issues

These tools support both operational needs and strategic security goals, forming a unified observability ecosystem across infrastructure, applications and network layers.

The [Secure by Design](#) approach is a government framework which embeds cyber security into digital services from the outset. Its goal is to make security an integral, proactive part of all government digital systems, rather than an afterthought.

Purpose

This document aims to explain how AWS's native monitoring tools align with the government's [Secure by Design](#) approach. It is not a technical how-to guide, but instead supports government organisations in understanding this alignment.

AWS monitoring tools include:

- [Amazon CloudWatch](#)
- [Amazon Inspector](#) and [Amazon GuardDuty](#)
- [AWS Config](#)
- [AWS Control Tower](#)
- [AWS CloudFormation](#)
- [AWS CloudTrail](#)
- [AWS Security Hub](#)
- [AWS Service Quotas](#) and [AWS Health Dashboard](#)
- [AWS Trusted Advisor](#)

By combining these, government organisations can implement a layered, automated and auditable monitoring framework. This approach not only strengthens the security posture, but also makes sure that operations align with Secure by Design principles, resulting in resilient AWS environments.

Secure by Design on the UK Government Security website

- [Secure by Design policy](#)
- [Implementing Secure by Design](#)
- [Guide to adopting Secure by Design](#)

Aligning AWS native monitoring tools with Secure by Design

Embedding monitoring within service design is a core tenet of secure cloud architecture. Monitoring is a key part of Secure by Design principle 5, [Build in detect and respond security](#). This is because being able to observe system behaviour and identify anomalies is crucial to detection and incident response.

By integrating monitoring early, organisations uphold principle 1, [Create responsibility for cyber security risk](#), so that operational owners, platform teams and security professionals have clear accountability for identifying and managing risks throughout the service life cycle.

AWS monitoring tools reinforce principle 8, [Defend in depth](#), by providing overlapping sources of telemetry metrics, logs, traces and event analysis. This means there are multiple layers of visibility across the environment. Monitoring becomes a cornerstone for governance, resilience and compliance, making it a central element of the [Managing observability](#).

AWS Shared Responsibility Model and monitoring

The [AWS Shared Responsibility Model](#) divides security and monitoring responsibilities between AWS and the customer.

AWS is responsible for securing and monitoring the underlying physical infrastructure, including:

- compute
- storage
- networking
- facilities

Customers, however, hold responsibility for securing and monitoring:

- workloads
- data
- configurations
- Identity and Access Management (IAM) policies
- applications running within the cloud environment

This separation supports Secure by Design principle 1, [Create responsibility for cyber security risk](#), by making it clear who is accountable. Platform teams, application teams and security functions each play a role in monitoring specific layers of the service. The model encourages government organisations to establish ownership structures so that monitoring responsibilities are understood, resourced and actively managed.

There's more about accountability in the Secure by Design activity [Agreeing roles and responsibilities](#).

AWS native monitoring ecosystem

The native monitoring ecosystem includes:

- Amazon CloudWatch for logs and metrics
- AWS CloudTrail for API auditing
- AWS Config for configuration visibility
- [AWS X-Ray](#) for distributed tracing
- specialised security monitoring services such as Amazon GuardDuty, Amazon Inspector and AWS Security Hub

This multi-layered approach meets Secure by Design principle 8, [Defend in depth](#). Each service covers different dimensions of observability (for example, focusing on behaviours, operations or security), so that if one layer misses an anomaly, another should detect it.

Native integration and service interoperability

Because AWS monitoring tools are natively integrated, telemetry can flow automatically between AWS services. For example:

- AWS CloudTrail logs can trigger Amazon CloudWatch Events
- Amazon GuardDuty findings can be aggregated in AWS Security Hub
- AWS Config can link compliance failures to [AWS Systems Manager](#) automation runbooks

This interoperability means that monitoring is not siloed, but contributes to a cohesive operational model.

Integrated workflows support Secure by Design principle 10, [Make changes securely](#), as monitoring outputs can initiate validation, remediation or rollback actions.

Native integrations also promote principle 4, [Design usable security controls](#), making things easier for operational teams by centralising visibility and streamlining investigations.

Native tools as secure-by-default components

AWS services are designed to operate securely with minimal configuration, but government organisations must still tailor monitoring to their risk profiles. Being able to enable encryption, retention policies, cross-account observability and automated remediation ensures that AWS native monitoring tools can support secure-by-default outcomes. This supports Secure by Design principle 7, [Minimise the attack surface](#), by putting secure configurations in place.

Amazon CloudWatch

[Amazon CloudWatch](#) is AWS's central observability platform, providing metrics, logs, events, traces, dashboards and anomaly detection for both AWS resources and customer applications.

CloudWatch supports Secure by Design principle 5, [Build in detect and respond security](#). It does so by enabling continuous detection of abnormal behaviour and initiating alerts or automated responses.

By offering comprehensive insights across performance, utilisation and operational trends, it supports principle 3, [Adopt a risk-driven approach](#) and principle 9, Embed continuous assurance, by allowing teams to validate system health over time.

Because it complements other AWS monitoring tools, which creates overlapping layers of visibility, it also supports principle 8, [Defend in depth](#).

Amazon CloudTrail

[Amazon CloudTrail](#) records API activity across the AWS environment. It is a foundational component for auditing, governance, compliance and security detection.

Because CloudTrail captures all control-plane activity and enables detection of suspicious or unauthorised actions, it aligns very well with Secure by Design principle 5, [Build in detect and respond security](#).

Amazon CloudTrail's control plane logs record management-level operations across AWS services, such as:

- creating or modifying IAM roles
- updating security group rules
- managing S3 bucket configurations
- starting and stopping EC2 instances

These logs provide visibility into who did what and when, supporting auditing, compliance and investigations into changes made to cloud resources.

Control plane logging does have some limitations. It does not capture detailed workload or data-access activity unless specific data events are enabled, and logs often arrive with some delay, limiting real-time detection. Coverage also varies across AWS services and high-volume logging can increase cost and complexity. For full observability, you should combine CloudTrail control plane logs with additional telemetry sources like VPC Flow Logs, CloudWatch Logs and GuardDuty.

CloudTrail's comprehensive log history means there's a traceable record of changes and actions for audits or investigations. This supports principle 9, [Embed continuous assurance](#).

AWS X-Ray

[AWS X-Ray](#) enables distributed tracing across:

- microservices
- container workloads
- serverless functions

It provides visibility into end-to-end execution paths, which highlights:

- bottlenecks
- dependency interactions
- anomalous behaviour

Tracing reveals latency spikes, unusual call patterns and errors across distributed systems, which supports Secure by Design principle 5, [Build in detect and respond security](#).

It also aligns with principle 3, [Adopt a risk-driven approach](#), because dependency insights inform threat modelling and architectural decisions.

And by offering deep visibility across both application and infrastructure components, AWS X-Ray's multi-layer view aligns with principle 8 ([Defend in depth](#)).

AWS Config

[AWS Config](#) is a configuration monitoring and compliance service. It continuously records changes to AWS resources and evaluates them against defined policies or best practices. By capturing resource states and relationships, AWS Config helps organisations maintain control over cloud environments while supporting Secure by Design principles.

AWS Config aligns with Secure by Design principle 9, [Embed continuous assurance](#), by enabling continuous validation of configuration compliance.

Because deviations from desired configurations can trigger alerts and remediation workflows, it also reinforces principle 5, [Build in detect and respond security](#).

And by allowing teams to focus monitoring and remediation efforts on high-value assets or critical configurations, it supports principle 3, [Adopt a risk-driven approach](#).

Amazon GuardDuty

[Amazon GuardDuty](#) is a managed threat detection service that continuously monitors for malicious or unauthorised behaviour. It uses machine learning, threat intelligence and anomaly detection to identify potential security incidents in real time.

By providing actionable findings that can trigger automated or human-led investigation, GuardDuty directly supports Secure by Design principle 5, [Build in detect and respond security](#).

By applying threat intelligence and analytics, it lets you prioritise high-risk threats, which aligns with principle 3, [Adopt a risk-driven approach](#).

Finally, its managed, automated nature supports principle 4, [Design usable security controls](#), because it can reduce operational overhead and improve usability for security teams.

AWS Security Hub

[AWS Security Hub](#) provides a centralised view of security findings from multiple AWS services, including Amazon GuardDuty, Amazon Inspector and AWS Config. It supports:

- compliance
- investigative workflows
- continuous security monitoring

Security Hub aligns with Secure by Design principle 9, [Embed continuous assurance](#), by consolidating findings and compliance checks into a single dashboard.

Because the aggregated insights mean you can identify and prioritise incidents more quickly, it reinforces principle 5, [Build in detect and respond security](#).

By centralising reporting and visibility, Security Hub supports principle 4, [Design usable security controls](#), and encourages accountability in line with principle 1, [Create responsibility for cyber security risk](#).

Amazon Inspector

[Amazon Inspector](#) is an automated vulnerability management service that assesses applications and workloads for security risks and deviations from best practices. Inspector helps proactively identify and prioritise vulnerabilities in both operating system and application layers.

Inspector can help your AWS deployment align closely with Secure by Design principle 3, [Adopt a risk-driven approach](#), by identifying the most critical security weaknesses and enabling remediation according to how severe the risk is.

It also supports principle 5, [Build in detect and respond security](#), by highlighting potential attack vectors before they are exploited.

And by providing actionable reports, you can make sure vulnerabilities are not only detected but efficiently addressed. This aligns with principle 4, [Design usable security controls](#).

AWS Trusted Advisor

[AWS Trusted Advisor](#) provides guidance to help you optimise AWS resources and improve security, performance and cost efficiency. It evaluates resources against AWS best practices and highlights misconfigurations or areas needing attention.

Trusted Advisor supports Secure by Design principle 3, [Adopt a risk-driven approach](#), by identifying high-impact operational and security risks.

By highlighting publicly exposed resources, overly permissive IAM roles and unencrypted storage, it helps you meet principle 7, [Minimise the attack surface](#).

And it also aligns with principle 4, [Design usable security controls](#), by providing actionable recommendations in an accessible format.

AWS CloudFormation drift detection

Using drift detection in [AWS CloudFormation](#) helps you identify configuration changes that deviate from your original infrastructure-as-code (IaC) templates. By detecting drift, you can make sure deployed resources remain consistent with their intended secure configurations.

Drift detection aligns with Secure by Design principle 9, [Embed continuous assurance](#), by continuously verifying the deployed environment against expected configurations.

Because it makes sure modifications are deliberate and controlled, it supports principle 10, [Make changes securely](#).

By identifying unexpected changes, drift detection also reduces the risk introduced by misconfigurations or unauthorised modifications. This aligns with principle 7, [Minimise attack surface](#).

AWS Control Tower

[AWS Control Tower](#) enforces best practices across accounts, simplifying governance, compliance and security management.

It provides a managed environment to automate the setup of secure multi-account AWS environments, which are known as [landing zones](#).

AWS Control Tower aligns with Secure by Design principle 1, [Create responsibility for cyber security risk](#), by providing a framework for accountability and governance.

Automated guardrails mean preventive and detective controls are embedded from the moment accounts are created. This supports principle 10, [Make changes securely](#), and principle 5, [Build in detect and respond security](#).

Continuous monitoring of guardrail compliance keeps accounts aligned with organisational security policies. This aligns with principle 9, [Embed continuous assurance](#).

AWS Service Quotas and Health Dashboard

Monitoring service limits and health status is critical for maintaining resilience and operational continuity. [AWS Service Quotas](#) and the [Health Dashboard](#) provide visibility into service usage, limits and potential disruptions.

This monitoring offers real-time insights into operational constraints and planned maintenance, which aligns with Secure by Design principle 9, [Embed continuous assurance](#).

Being aware of service limits also supports principle 3, [Adopt a risk-driven approach](#). This lets you plan capacity while reducing the likelihood of outages that could impact security or compliance.

More information

- [Secure by Design Principles \(UK Government Security\)](#)
- [Secure by Design Activities \(UK Government Security\)](#)
- [Data Protection Act 2018 \(Legislation.gov.uk\)](#)
- [The cloud security principles \(NCSC\)](#)